

ERIC GABRIEL MANABAT

FULL-STACK AI ENGINEER

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EDUCATION

Pamantasan ng Lungsod ng Valenzuela | *Bachelor of Science in Information Technology*

2023 - 2027

Honors: **Consistent Dean's Lister (1.03 Running GWA)** | Valenzuela City, Philippines

PROFESSIONAL EXPERIENCE

FlyRank AI | *Full-Stack AI Engineer Intern*

Jun 2026 - Present | Remote

- **Accomplished** an increase in retrieval accuracy from **81.8% to 86.4% Top-1** on ground-truth benchmarks by engineering an autonomous FastAPI AI sidecar with hybrid BM25 + dense vector search, Reciprocal Rank Fusion (RRF k=60), and deterministic blend reranking.
- **Accomplished** elimination of out-of-domain hallucinations (**100% negative query pass rate** and **90% LLM-as-judge relevance score**) by designing a bounded 3-step **LangChain** agentic query loop with cosine similarity gating (<0.50 threshold) and inline citation grounding.
- **Accomplished** a **~65% reduction** in time-to-first-token (TTFT) and established real-time observability by implementing asynchronous Server-Sent Events (SSE) token streaming APIs, Prometheus /*metrics* latency histograms, and Grafana telemetry dashboards.
- **Accomplished** end-to-end user-facing AI delivery by progressing from backend microservices to developing real-time frontend streaming chat interfaces with automated test-driven evaluation pipelines.

Nexvision Innovations Inc. | *Full Stack Software Engineering Intern - Team Lead*

Jun 2026 - Present | Marikina (Hybrid)

- **Accomplished** 100% compliance with Department of Labor and Employment (DOLE) statutory standards across 4 enterprise applications by designing interval-partitioned time algorithms to isolate 10 PM - 6 AM night differentials from standard overtime multipliers.
- **Accomplished** automated statutory deduction and 13th-month proration calculations across **500+ employee records** by developing modular TypeScript/Next.js calculation engines with immutable audit logging.
- **Accomplished** reduction of monthly branch audit preparation time from **3 days to under 5 minutes** by building high-throughput multi-sheet Excel/CSV data export engines and state persistence modules.
- **Accomplished** a **40% increase in sprint execution velocity** across 180+ Jira/GitHub issues as Intern Team Lead by delegating frontend/backend tasks, conducting peer code reviews for TypeScript type safety, and leading architectural oversight.

FEATURED AI PROJECTS

PLV eLib - Academic Library AI Sidecar | *FastAPI, Laravel, LangChain, PostgreSQL, SQLite FTS5, Prometheus, Grafana*

2026

- **Accomplished** high-precision semantic resource discovery (lifting Top-1 accuracy from **81.8% to 86.4%**) by engineering a decoupled FastAPI AI sidecar fusing SQLite FTS5 (BM25) and dense embeddings via Reciprocal Rank Fusion (RRF k=60).
- **Accomplished** zero-hallucination factual grounding (**100% negative query pass rate**) by building a bounded **LangChain** agentic retrieval loop with deterministic refusals on out-of-domain queries and automated LLM-as-judge verification.
- **Accomplished** real-time telemetry and query observability by configuring Prometheus latency histograms and a 6-chart Dockerized Grafana dashboard to track production inference latency and live feedback streams.

AGOS - AI-Guided Overflow Surveillance | *FastAPI, React, YOLOv8, OpenCV, Docker, Raspberry Pi*

2026

- **Accomplished** real-time floating debris and waterway blockage detection by fine-tuning a custom **YOLOv8** object detection model on **5,000 CVAT-annotated** drainage images.
- **Accomplished** high-accuracy frame ingestion and noise reduction under variable weather conditions by engineering an automated **OpenCV** preprocessing pipeline to filter motion blur and eliminate glare.
- **Accomplished** a **92% reduction in cloud GPU compute overhead** by architecting a decoupled batch inference engine that evaluated camera feeds on an optimized 3-minute inference cadence.
- **Accomplished** automated emergency flood warnings with zero duplicate alerts by developing a multi-modal decision fusion engine (fusing visual blockage, ultrasonic depth, and weather APIs) with a 30-minute deduplication cooldown.

TECHNICAL SKILLS

AI & Machine Learning: LangChain, LangGraph, RAG Pipelines, Vector Search (pgvector, Hybrid BM25/Dense RRF), Embeddings, YOLOv8, OpenCV, LLM-as-Judge Evaluation, Prompt Engineering, SSE Streaming, Prometheus & Grafana Observability

Languages: Python, TypeScript, JavaScript, SQL, C#, Java, PHP

Frameworks & Full-Stack: FastAPI, Next.js, React, Node.js, Laravel, Tailwind CSS

Databases & Cloud: PostgreSQL, Redis, Supabase, Docker, AWS, GitHub Actions

HONORS & CERTIFICATIONS

- **LLM Zoomcamp - Production LLM & RAG Engineering Bootcamp** | DataTalks.Club | Aug 2026
- **Learn RAG (Retrieval-Augmented Generation)** | Boot.dev | 2026
- **Global Cyber Skills Benchmark (Project Nightfall)** - Ranked 63/589 Global Teams | HackTheBox | May 2026
- **Google Cybersecurity Professional Certificate** | Google | Jan 2026
- **Mastering REST APIs with FastAPI** | Packt / Coursera | May 2026
- **ITLympics 2026 - Silver Medal** | Capture the Flag | Pamantasan ng Lungsod ng Valenzuela | Mar 2026